



AcePharm

INTERNAL QA DOCUMENT — CONFIDENTIAL

QA Checklist & User Manual

Step-by-step manual testing guide covering the Marketing Site, Learner App, Admin Panel, and API layer of the AcePharm platform.

apps/marketing

apps/app

apps/api

Admin Panel

Cross-Browser & A11y

PROJECT

AcePharm UK Revision Platform

VERSION

1.0 — Sprint Release

DATE

September 2026

TOTAL TEST CASES

98 manual test cases

Module Overview

46

Marketing Site Tests

30

Learner App Tests

9

Admin Panel Tests

8

API Layer Tests

5

Cross-Browser & A11y

98

Total Test Cases

▲ IMPORTANT — TEST ENVIRONMENT URLS

- **MARKETING SITE:**

<https://acepharmexams.co.uk>

- **LEARNER APP:**

<https://app.acepharmexams.co.uk>

- **BACKEND API:**

<https://acepharm-api.takweencentreuk.workers.dev>

Complete all checklists sequentially. Use a fresh **INCOGNITO/PRIVATE BROWSER WINDOW**

for each major section.

MODULE A — Marketing Site apps/marketing

A1 · Homepage (/)

Prerequisites: Open <https://acepharmexams.co.uk> in a fresh incognito window.

#	Test Step	Expected Result	☐
A1.1	Load the homepage	Page renders with AcePharm logo in header, gradient hero section, and no console errors	☐
A1.2	Verify hero section heading	Title reads "Pharmacy revision that shows you exactly what to study next."	☐
A1.3	Observe Interactive Sample Question widget loading	A question card appears labelled ACP-CV-0012 in Cardiovascular category	☐
A1.4	Read the clinical vignette stem	Full patient scenario visible (Afro-Caribbean male, BP 154/94 mmHg, normal eGFR)	☐
A1.5	Click any answer option without submitting	Selected option highlights with indigo border ring; no other options change	☐
A1.6	Click "Cover Options (Active Recall)" button	All 5 options hide; dashed box with "Options covered for diagnostic recall" appears	☐
A1.7	Click "Click to uncover options"	All 5 options reappear; previous selection is restored	☐
A1.8	Select option A ("Ramipril") and click Submit Answer	Options lock; Ramipril shows red/rose border (incorrect); Amlodipine shows teal (correct). Each option shows rationale	☐
A1.9	Verify correct rationale for Option B	Displays: "NICE NG136... Step 1 antihypertensive... Black African or African-Caribbean origin... CCB"	☐
A1.10	Verify NICE Key Takeaway card appears	Contains guideline reference and CTA: "Start Full Free Revision Bank →"	☐
A1.11	Click Reset Demo	Widget resets — all options deselected, Submit button greyed out	☐
A1.12	Select correct Option B and submit	 emoji visible, green correct highlight, rationales shown	☐
A1.13	Click "Start Full Free Revision Bank →" CTA	Navigates to https://app.acepharmexams.co.uk/auth/register	☐
A1.14	Click hero "Start revising free" button	Navigates to the registration page	☐

#	Test Step	Expected Result	☐
A1.15	Click " Explore the platform " button	Navigates to <code>/features</code>	☐
A1.16	Test mobile responsive layout at 375px (iPhone SE)	Question card readable; options stacked vertically; buttons full-width	☐

A2 · Question Bank Page (`/question-bank`)

#	Test Step	Expected Result	☐
A2.1	Navigate to <code>/question-bank</code>	Page loads with hero, interactive question widget, pathway grid, and syllabus explorer	☐
A2.2	Verify sample question widget in hero	Same <code>ACP-CV-0012</code> widget with all features (same as A1.3–A1.12)	☐
A2.3	Scroll to "Choose your pathway" section	5 pathway cards. "MPharm revision" has Active teal badge; 4 others show "Coming later"	☐
A2.4	Scroll to GPhC Therapeutic Categories section	Heading reads: "Explore GPhC Therapeutic Categories" with subtitle and search bar	☐
A2.5	Count initial category cards	19 cards; stat reads "Showing 19 of 19 GPhC curriculum categories"	☐
A2.6	Type " cardio " in search box	Real-time filter to "Cardiovascular System" only; stat changes to "Showing 1 of 19"	☐
A2.7	Type " DOAC " in search box	Filters to "Cardiovascular System" (DOAC Dosing is a key topic); 1 result	☐
A2.8	Type " MEP " in search box	Filters to "Pharmacy Law, Ethics & Practice"	☐
A2.9	Clear search; filter by " High " weighting pill	Shows only High-weight categories: CV, Resp, CNS, Infection, Endocrine, Calculations, Law, Paediatrics, Elderly, OTC	☐
A2.10	Filter by " Low " weighting pill	Shows Eye, ENT, Anaesthesia — 3 low-weight categories	☐
A2.11	Hover over a category card	Border transitions to indigo; title turns indigo — hover effect applied	☐
A2.12	Click " Practice Topic → " on any card	Navigates to <code>https://app.acepharmexams.co.uk/auth/register</code>	☐
A2.13	Type " xyzabc " (nonsense term)	Empty state: "No categories match 'xyzabc'" with a Clear Filters button	☐
A2.14		Search resets; all 19 categories reappear	☐

#	Test Step	Expected Result	☐
	Click Clear Filters in empty state		

A3 · Pricing Page (/pricing)

#	Test Step	Expected Result	☐
A3.1	Navigate to /pricing	Page loads with hero, PricingToggle island, PlanComparisonMatrix island	☐
A3.2	Verify default billing state	Annual billing toggled ON; Yearly card highlighted with indigo ring; Monthly dimmed	☐
A3.3	Toggle to Monthly billing	Monthly card gets indigo ring; Yearly drops to outline border	☐
A3.4	Scroll to Student Discount banner	Teal-indigo gradient card with STUDENT20 mention and promo input field	☐
A3.5	Enter student20 (lowercase) and click Apply	Code normalises to uppercase; success banner: "20% discount will apply at checkout (£39.99/year)"	☐
A3.6	Click Apply with empty field	Nothing happens; no error message displayed	☐
A3.7	Enter INVALID123 and click Apply	Error: "Invalid coupon code. Try STUDENT20 or BPSA."	☐
A3.8	Enter BPSA and click Apply	Success banner appears (identical to STUDENT20 behaviour)	☐
A3.9	Verify Feature Comparison Matrix header	Reads "Feature Comparison Matrix"; collapse toggle button visible	☐
A3.10	Click "Collapse Table ▲"	Table body hides; button label changes to "Expand Table ▼"	☐
A3.11	Click "Expand Table ▼"	Table body re-appears with all rows	☐
A3.12	Verify table section headings	Three category groups: Question Bank & Mocks, AI Tutoring & Coaching, Spaced Repetition & Analytics	☐
A3.13	Check Free column for "Ace AI 24/7 Tutor" row	Shows × (cross / not available)	☐
A3.14	Check Pro column for same row	Shows "Unlimited inquiries & step explanations" in indigo	☐
A3.15	Check "Practice Questions Monthly Access" row	Free: "30 questions / month"; Pro: "Unlimited (1,500+ items)"	☐
A3.16		Card visible with "Enquire about university licensing →" link	

#	Test Step	Expected Result	☐
	Scroll to University & Cohort section		☐
A3.17	Click university enquiry link	Navigates to <code>/contact?subject=institutional</code>	☐

A4 · FAQ Page (`/faq`)

#	Test Step	Expected Result	☐
A4.1	Navigate to <code>/faq</code>	Page loads with three accordion groups and all questions visible	☐
A4.2	View page source → search <code>application/ld+json</code>	FAQPage JSON-LD schema present with all 11 questions in <code>mainEntity</code> array	☐
A4.3	Click any question accordion item	Expands smoothly to reveal answer text	☐
A4.4	Click a second question	Second expands; verify consistent group interaction behaviour	☐

A5 · SEO, Sitemap & AI Crawling Directives

#	URL / Step	Expected Result	☐
A5.1	Open <code>/robots.txt</code>	User-agent: * allows <code>/question-bank</code> , <code>/pricing</code> , <code>/faq</code> ; Disallows <code>/app/</code> , <code>/admin/</code> , <code>/api/</code>	☐
A5.2	Verify AI bot directives in robots.txt	Named entries for <code>GPTBot</code> , <code>ClaudeBot</code> , <code>PerplexityBot</code> , <code>Google-Extended</code> all with <code>Allow: /</code>	☐
A5.3	Open <code>/sitemap.xml</code>	Valid XML sitemap with <code>/</code> , <code>/features</code> , <code>/pricing</code> , <code>/question-bank</code> , <code>/faq</code> , and <code>/blog/*</code> entries	☐
A5.4	Open <code>/llms.txt</code>	Markdown file loads with AcePharm description, resource URLs, and clinical invariants	☐
A5.5	View homepage source — search <code>ld+json</code>	<code>@graph</code> array contains <code>Organization</code> , <code>EducationalOccupationalProgram</code> , and <code>WebSite</code> types	☐
A5.6	Verify <code>AggregateRating</code> in LD+JSON	<code>ratingValue: "4.9"</code> , <code>ratingCount: "342"</code> inside <code>EducationalOccupationalProgram</code>	☐

MODULE B — Learner App apps/app

B1 · Registration & Branded Email Verification

Prerequisites: Use a fresh email (e.g. `yourname+test1@gmail.com`). Open `https://app.acepharmexams.co.uk/auth/register`.

#	Test Step	Expected Result	☐
B1.1	Open <code>/auth/register</code>	Registration form loads with name, email, and password fields; no console errors	☐
B1.2	Fill in name, email, password (min 8 chars)	Form accepts input; no premature validation errors	☐
B1.3	Submit the registration form	Success state shown; notification that a branded verification email has been sent	☐
B1.4	Check inbox for verification email	Email arrives from AcePharm (not Firebase noreply); branded AcePharm header present	☐
B1.5	Inspect email HTML visually	Contains AcePharm branding, pharmacist context, green "Verify Email" button	☐
B1.6	Click the "Verify Email" button in email	Opens <code>https://app.acepharmexams.co.uk/auth/action?mode=verifyEmail&oobCode=...</code>	☐
B1.7	Wait 2–3 seconds on action page	Verification success state shown; auto-redirects to <code>/session/new</code> or dashboard	☐

B2 · Login & Forgot Password Flow

Prerequisites: Use the account registered in B1.

#	Test Step	Expected Result	☐
B2.1	Open <code>/auth/login</code>	Login page with email/password fields and "Forgot Password?" link visible	☐
B2.2	Enter wrong password and submit	Branded error message shown; no Firebase-native dialog	☐
B2.3	Click "Forgot Password?"	In-page panel slides/reveals with an email input; no redirect to Firebase	☐
B2.4		Email field accepts input	☐

#	Test Step	Expected Result	☐
	Enter registered email in forgot-password panel		
B2.5	Click "Send Reset Email"	Confirmation state: "Check your inbox" with 60-second countdown timer visible	☐
B2.6	Observe countdown timer	Timer counts down every second: "Resend in 59s", "Resend in 58s"...	☐
B2.7	After 60 seconds, verify Resend button re-enables	Button becomes active; clicking resets the 60s timer and queues another email	☐
B2.8	Check inbox for password reset email	Branded AcePharm email arrives; not a Firebase default email	☐
B2.9	Inspect the reset email body	AcePharm branding, request metadata, red "Reset Password" CTA button	☐
B2.10	Click "Reset Password" button in email	Opens <code>/auth/action?mode=resetPassword&oobCode=...</code>	☐
B2.11	Verify password strength meter renders	Real-time entropy bar and requirement checklist visible on page	☐
B2.12	Type <code>password</code> (weak)	Meter shows Weak/Low; missing Uppercase, Number requirements highlighted in red	☐
B2.13	Type <code>Acepharm2024!</code> (strong)	Meter shows Strong/High; all requirements ticked (8+ chars, uppercase, lowercase, number)	☐
B2.14	Submit the strong new password	Success state; auto-logged in and redirected to <code>/session/new</code> or dashboard	☐
B2.15	Login with new password on <code>/auth/login</code>	Successful login; dashboard content visible	☐

B3 · Dashboard, Route Guards & Error Pages

#	Test Step	Expected Result	☐
B3.1	Log in successfully	Dashboard (<code>/</code>) loads showing session stats, question count, streak	☐
B3.2	Open DevTools → Application → SessionStorage	No sensitive credentials stored; only session-related keys present	☐
B3.3	Access <code>/admin</code> as regular user	"Access Restricted" card shown; admin content never loads	☐

#	Test Step	Expected Result	☐
B3.4	Log out	Redirected to <code>/auth/login</code>	☐
B3.5	Access <code>/</code> while logged out	Redirected to <code>/auth/login</code> ; dashboard content not visible	☐
B3.6	Visit <code>/does-not-exist</code> (any bad URL)	Custom 404 page loads with AcePharm branding and "Return to Dashboard" button	☐
B3.7	Trigger a JavaScript runtime error or visit error boundary	Custom error page renders with "Try Again" recovery action — not a browser error screen	☐

B4 · Session — Question Player & State Persistence (SessionStorage)

Prerequisites: Log in as verified learner → navigate to `/session/new` → start a session.

#	Test Step	Expected Result	☐
B4.1	Start a new session from <code>/session/new</code>	Question player at <code>/session/active/[id]</code> loads with a clinical question and 5 options	☐
B4.2	Click any answer option	Option highlights; no premature answer reveal	☐
B4.3	Check DevTools → Application → SessionStorage	Key <code>acepharm:session:[id]:q:[qId]</code> present with <code>selectedOptionId</code> stored	☐
B4.4	Type text into personal notes field (if visible)	Notes field accepts input	☐
B4.5	Check SessionStorage again after typing	<code>draftNotes</code> key stores current text in real-time	☐
B4.6	Refresh the page (F5 / Cmd+R)	Page reloads and automatically restores the previously selected option and typed notes	☐
B4.7	Navigate back then forward in the browser	Same selections restored after navigation	☐
B4.8	If timer mode enabled, note the timer value then refresh	Timer state restores from SessionStorage — not reset to zero	☐
B4.9	Submit answer and verify rationale reveal	Correct answer highlighted in green; incorrect options marked; full rationale visible	☐
B4.10	Complete the session	Session summary at <code>/session/summary/[id]</code> loads with score and analytics breakdown	☐

MODULE C — Admin Panel

[apps/app/admin](#)

Prerequisites: Log in with an account that has `isAdmin: true` on its Firestore profile. Navigate to <https://app.acepharmexams.co.uk/admin>.

#	Test Step	Expected Result	☐
C1	Log in as admin → navigate to /admin	Admin layout loads with sidebar navigation; no redirect occurs	☐
C2	Navigate to /admin/questions	Questions management page loads with data	☐
C3	Navigate to /admin/review	Review queue page loads (requires reviewer/admin role)	☐
C4	Navigate to /admin/users	User management list loads with learner accounts	☐
C5	Navigate to /admin/curriculum	Curriculum mapping management page loads	☐
C6	Navigate to /admin/blog	Blog content management loads correctly	☐
C7	Navigate to /admin/audit	Audit log table loads with entries	☐
C8	Log out → attempt to access /admin directly	Redirected to login page — no admin content exposed	☐
C9	Log in as regular learner → visit /admin URL	"Access Restricted" card renders; no admin data or routes accessible	☐

MODULE D — API Layer apps/api

Tool: Use `curl`, Postman, or browser DevTools Network tab to verify these endpoints against `https://acepharm-api.takweencentreuk.workers.dev`.

D1 · Branded Auth Email Endpoints

#	Endpoint	Method	Test Payload	Expected Response	☐
D1.1	<code>/api/v1/auth/send-custom-verification</code>	POST	<code>{ "email": "test@example.com", "uid": "...", }</code>	<code>200 { success: true }</code> — branded verification email dispatched	☐
D1.2	<code>/api/v1/auth/send-custom-password-reset</code>	POST	<code>{ "email": "test@example.com" }</code>	<code>200 { success: true }</code> — branded reset email dispatched	☐
D1.3	<code>/api/v1/auth/send-custom-password-reset</code>	POST	<code>{ "email": "notexist@example.com" }</code>	<code>404</code> or safe error — no user found	☐
D1.4	<code>/api/v1/auth/send-custom-verification</code>	POST	Empty body <code>{}</code>	<code>400</code> validation error	☐
D1.5	Any authenticated route	GET	No <code>Authorization</code> header	<code>401 Unauthorized</code>	☐

D2 · General API Health

#	Test Step	Expected Result	☐
D2.1	<code>GET /api/v1/questions</code> with valid auth token	Returns question list JSON with status 200	☐
D2.2	<code>GET /api/v1/sessions</code> with valid auth token	Returns session history for the authenticated user	☐
D2.3	Validate CORS headers on API responses	<code>Access-Control-Allow-Origin</code> present for <code>app.acepharmexams.co.uk</code>	☐

MODULE E — Cross-Browser & Accessibility

#	Test	Browsers	Expected Result	☐
E1	Homepage & Interactive Question Widget full render	Chrome 127, Firefox 128, Safari 17, Edge	No visual regressions; widget fully interactive across all browsers	☐
E2	Pricing page comparison matrix table	Chrome + Firefox	Table renders correctly; alternating row colours; no overflow	☐
E3	Question player keyboard navigation	Chrome	Tab → options; Enter → select; Tab → Submit; Enter → submit answer	☐
E4	Colour contrast — option feedback colours	Any browser	Green (teal) correct / Red (rose) incorrect pass WCAG AA 4.5:1 contrast ratio	☐
E5	Screen reader option state announcements	VoiceOver (macOS)	<code>role="radio"</code> <code>aria-checked</code> state correctly announced when option selected	☐

Bug Reporting Reference

When raising a defect in the QA tracker, include the following fields:

Title:

[Module] - [Short Description]

URL:

[Full URL where the bug occurs]

Steps to Reproduce:

1. ...

2. ...

3. ...

Expected:

[What should happen]

Actual:

[What happened instead]

Browser:

[e.g. Chrome 127 / Safari 17 / Firefox 128]

Severity:

Critical | High | Medium | Low

Screenshot/Video:

[Attach if visual defect]

Severity Definitions

Severity	Definition	Example
Critical	Blocks core flow; no workaround	Cannot register, email not sent, login fails
High	Feature broken; workaround exists	SessionStorage not restoring options on refresh
Medium	Feature degraded; usable with effort	Promo code error message not displaying

Severity	Definition	Example
Low	Cosmetic or minor UX issue	Hover colour slightly off on category card

5-Minute Smoke Test Shortlist

PRE-RELEASE SANITY CHECKS

Run before any production deployment. These 8 checks cover the most critical user-facing features.

1. **A1.3–A1.12** — Interactive Sample Question demo (select, cover, submit, rationale, reset)
2. **A2.6–A2.8** — Syllabus Explorer real-time search filtering
3. **A3.5** — Promo code `STUDENT20` applies successfully
4. **A3.10–A3.11** — Comparison matrix expand/collapse toggle
5. **B2.3–B2.6** — In-page Forgot Password panel with 60s countdown
6. **B2.11–B2.13** — Password strength meter on reset page
7. **B4.6** — Session state persists across page refresh
8. **B3.6** — Custom 404 page renders on bad URL